Language research and the "Cognitive Revolution"

Lecture 6

Language as a formal/symbolic system

There is a very old tradition of describing natural languages in abstract, formal terms similar to those used in Artificial Intelligence.

Experimental psychology in the 50's and 60's was actively hostile to talking about this kind of "internal state."

Skinner's *Verbal Behavior* tried to describe how language could be learned by operant conditioning.

Chomsky's (1959) review of *Verbal Behavior* is better remembered than the book itself.

This was an important moment in "The Cognitive Revolution"

Where does this tradition come from?

Interest in the concept of "Artificial Intelligence" dates back to the beginning of the digital computer.

Alan Turing, who developed decryption devices for the British Army during WWII, thought about this a lot.

Basically, computers were invented as "universal function approximators." No one had really made machines that did so many different kinds of calculations and logical operations before. This raised the question of how the computer differed from human intelligence, if at all.



The “Turing Test”

Original paper is here: http://cogprints.org/499/1/turing.html



The imitation game:
A and B are a man and a woman, C is an “examiner.”
C’s job is to figure out the gender of A and B.
A is trying to fool C.
B is trying to help C guess the correct answer.

The test:
IF you can replace either A or B with a computer, and C doesn’t notice, the computer has demonstrated “intelligence.”

Some Interesting features of the “Turing Test”

- role of natural language

- deception

- Turing explicitly excludes "expert" behaviors like calculating large sums or composing poetry

- Turing concludes that the best way to get an intelligent machine would be to build something with the cognitive capabilities of a child, and have it learn to be more like an adult.

A later example of GOFAI:
Newell & Simon's General Problem Solver

Given an abstract characterization of a formal problem, apply procedures to symbols to find a solution.

Very good at games/puzzles:

- Tower of Hanoi
- Chess

Also good at mathematics/logic:

- geometry proofs
- word problems (algebra)

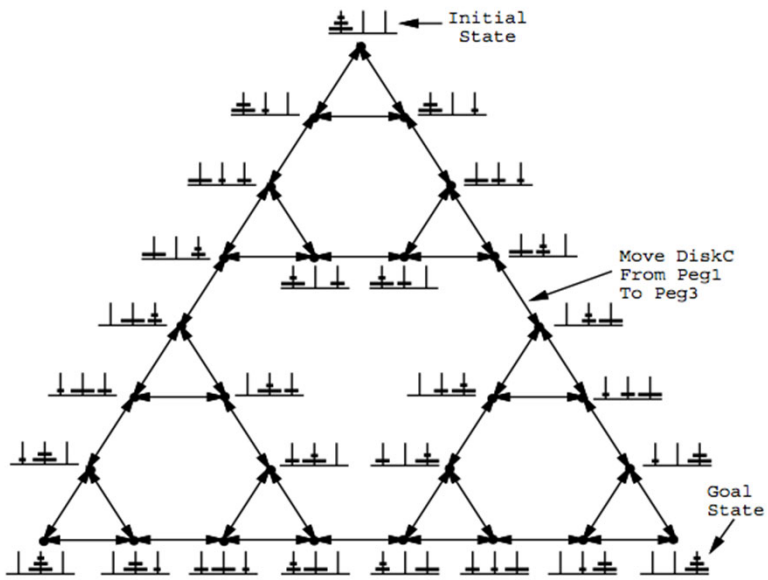


Figure 2: State Space for the Three-Disk Tower of Hanoi Puzzle

Figure from "Abstracting the Tower of Hanoi" (Knoblock, 1990)

Why the emphasis on formal problems?

Technology?

–Inputs to computers still very limited (punchcards, machine language, etc.)

–No way to input a high-resolution image and say “find the face,” for example…

Theoretical commitments:

–Intelligence is *exemplified* by this kind of rational behavior

–You can train a pigeon to guide a missile, but it doesn’t know what it’s doing.

–(There’s a deep argument about whether AI programs do, either, but they at least represent goal states and choose procedures for achieving them.)

Some highlights of Chomsky's critique

The construction of a grammar which enumerates sentences in such a way that a meaningful structural description can be determined for each sentence … *characterizes* abstractly the ability of one who has mastered the language to *distinguish sentences from nonsentences*, to understand new sentences (in part), to note certain ambiguities, etc. These are very remarkable abilities.

It is easy to show that the new events that we accept and understand as sentences *are not related to those with which we are familiar by any simple notion of formal (or semantic or statistical) similarity* or identity of grammatical frame. Talk of generalization in this case is entirely pointless and empty.

It appears that we recognize a new item as a sentence not because it matches some familiar item in any simple way, but because it is generated by the grammar that each individual has somehow and in some form internalized. And we understand a new sentence, in part, because we are somehow capable of determining the process by which this sentence is derived in this grammar.

The fact that all normal children acquire essentially comparable grammars of great complexity with remarkable rapidity suggests that human beings are somehow *specially designed to do this*, with *data-handling or "hypothesis-formulating" ability* of unknown character and complexity.

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Translation:

Linguistic knowledge is characterized as a grammar: *a set of abstract rules that produce grammatical sentences.*

There are an infinite number of grammatical sentences, which makes the fact that we can make a grammaticality judgment for any new sentence remarkable.

People's intuitions about whether sentences are grammatical are the key data for the study of language.

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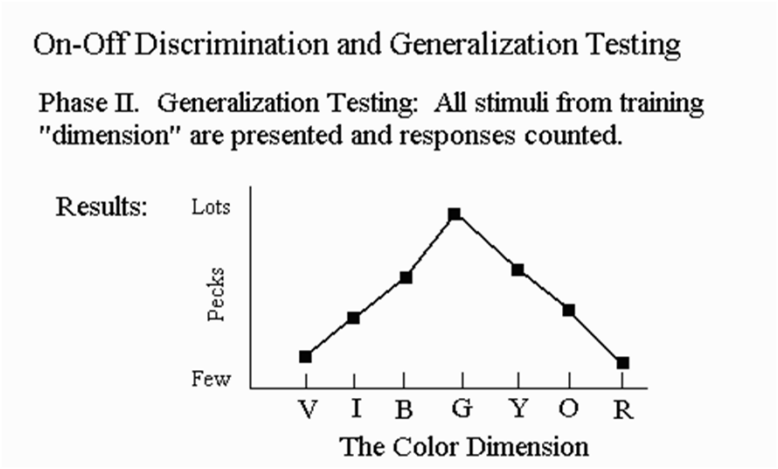
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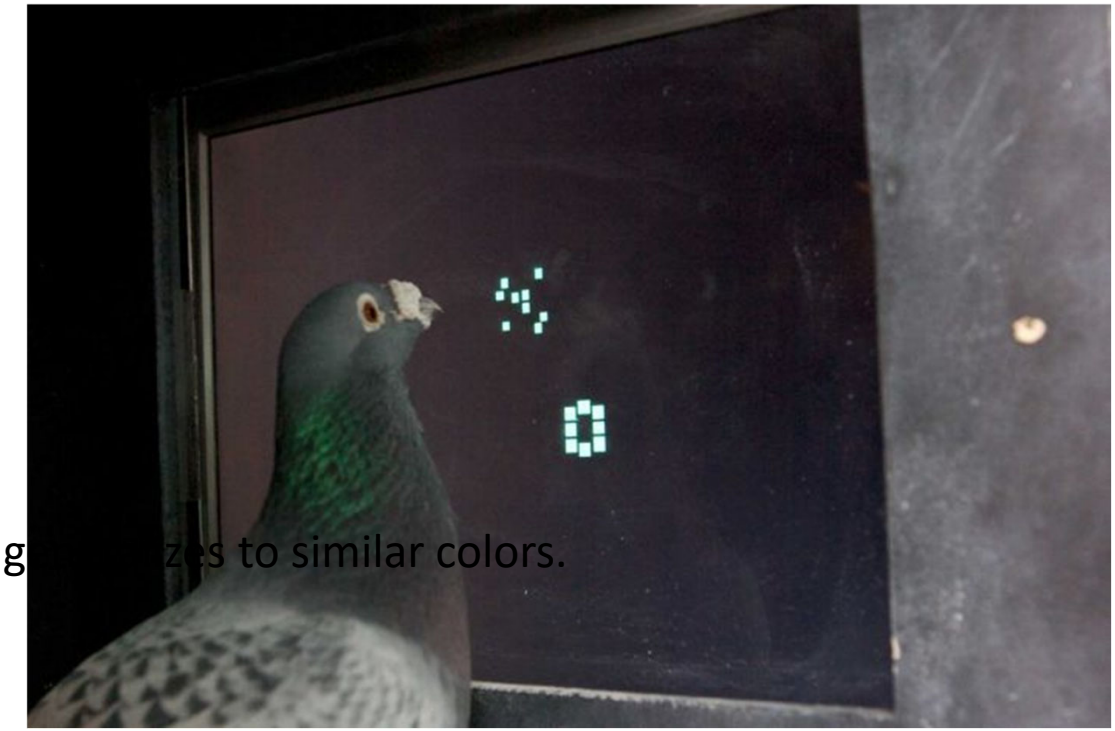
Grammatical sentences don't have any obvious similarities to one another.

Therefore, the behaviorist concept of "stimulus generalization" can't deal with the infinite number of sentences a grammar can generate.

Stimulus generation in behaviorism



Trained to peck for a *green* light, responding g zes to similar colors.



This pigeon is doing a magnitude discrimination task.

Cases in which operant conditioning generalizes depend on relatively *concrete* similarity:
- color
- number, etc.

It's not clear how this extends to abstract concepts.

Stimulus control

A typical example of stimulus control for Skinner would be the response to a piece of music with the utterance “M

Other examples of stimulus control merely add to the general mystification. Thus, a proper noun is held to be a res

Stimulus control

“We cannot predict verbal behavior in terms of the stimuli in the speaker's environment, since we do not know what the current stimuli are until he responds.”

Interpretation:

Skinner doesn't just have a bad theory of language. He is posing the wrong question.

You can't hope to explain why people say what they do, when they do. That's determined by a whole matrix of social interactions, cognitive processes, previous experiences, etc., much of which has nothing to do with language.

The tractable problem is to study the structure of what people say, and from there try to derive the set of rules that produces this structure.

Colorless green ideas sleep furiously

“Neither (a) 'colorless green ideas sleep furiously' nor (b) 'furiously sleep ideas green colorless', nor any of their parts, has ever occurred in the past linguistic experience of an English speaker. But (a) is grammatical, while (b) is not.” (Chomsky, 1955)

We might add further that “Car! Getting towed! Yours!” is ungrammatical, though the 'stimulus' that likely 'controlled' it is perfectly clear.

Chomsky is trying to describe language at the "computational level of analysis."

This term comes from David Marr, a visual neuroscientist who talked about cognitive systems as having three "levels of analysis:"

1. Computational
2. Algorithmic
3. Implementational

An example will help here…

Cash registers



Computational level of analysis: given prices of items, determine amount due for a purchase; given payment amount, compute amount due as change.



Algorithmic level: use operations of addition (price) and subtraction (change).

Implementational level: Doesn't matter! As far as these functions are concerned, a credit card reader or a computer program are basically "cash registers."



This example borrows heavily from a lecture by Sharon Goldwater, worth looking at in its own right: http://www.inf.ed.ac.uk/teaching/courses/ccs/lectures/ccs03_marrACT-2x2.pdf

Type I problems

Marr referred to problems that break down cleanly in these terms as “Type I” problems.

Viewing language as a Type I problem means you can work at the computational (Universal Grammar) and algorithmic level (specific processes and how they’re applied in different languages) *without worrying about the implementational level* (errors, memory limitations, associative learning, etc.)

A note on chronology: Chomsky’s critique pre-dates Marr’s theoretical work by about a decade. Both were influential in artificial intelligence research and theoretical work in the GOFAI tradition.

Another way to think about this: Marr's "Type 2" problems

Chomsky's [2] notion of a "competence" theory for English syntax is precisely what I mean by a computational theory for that problem. Both have the quality of being little concerned with the gory details of algorithms that must be run to express the competence (i.e. to implement the computation). That is not to say that devising suitable algorithms will be easy, but it is to say that before one can devise them, one has to know what exactly it is that they are supposed to be doing, and this information is captured by the computational theory. When a problem decomposes in this way, I shall refer to it as having a *Type 1* theory.

The fly in the ointment is that while many problems of biological information processing have a Type 1 theory, there is no reason why they should all have. This can happen when a problem is solved by the simultaneous action of a considerable number of processes, *whose interaction is its own simplest description*, and I shall refer to such a situation as a *Type 2* theory.[3]

Another way to think about this: Marr's "Type 2" problems

The reason for drawing this point out at such length is that it bears upon another issue, namely the type of theory that the grammar of natural language might have. The purpose of human language is presumably to transform a datastructure that is not inherently one-dimensional into one-dimensional form for transmission as a sequential utterance, thereafter to be retranslated into some rough copy of the original in the head of the listener. Viewed in this light, it becomes entirely possible that there may exist no Type 1 theory of English syntax of the type that transformational grammar attempts to define—that its constraints resemble wired-in conventions about useful ways of executing this tedious but vital operation, rather than deep principles about the nature of intelligence. An abstract theory of syntax may be an illusion, approximating what really happens only in the sense that Julesz's second order statistics theory approximates the behaviour of the set of processes that implement texture vision and which, in the final analysis, are all the theory that there is. In other words, the grammar of natural language may have a theory of Type 2 rather than of Type 1.

In other words…

Details about how language is implemented in the brain might actually be unavoidable in a theory of language.

There is actually a whole alternative approach to pursuing language research based on just this hypothesis.

Emergentism/Statistical Learning/Parallel Distributed processing

The strong commitments on this view are at the *implementational level:*

-distributed (non-hierarchical) representations (like activity patterns over groups neurons)

-gradual learning

(like adjustment of the strength of synaptic connections)

- focus on "microstructure" and more *graded* division between "levels of analysis”

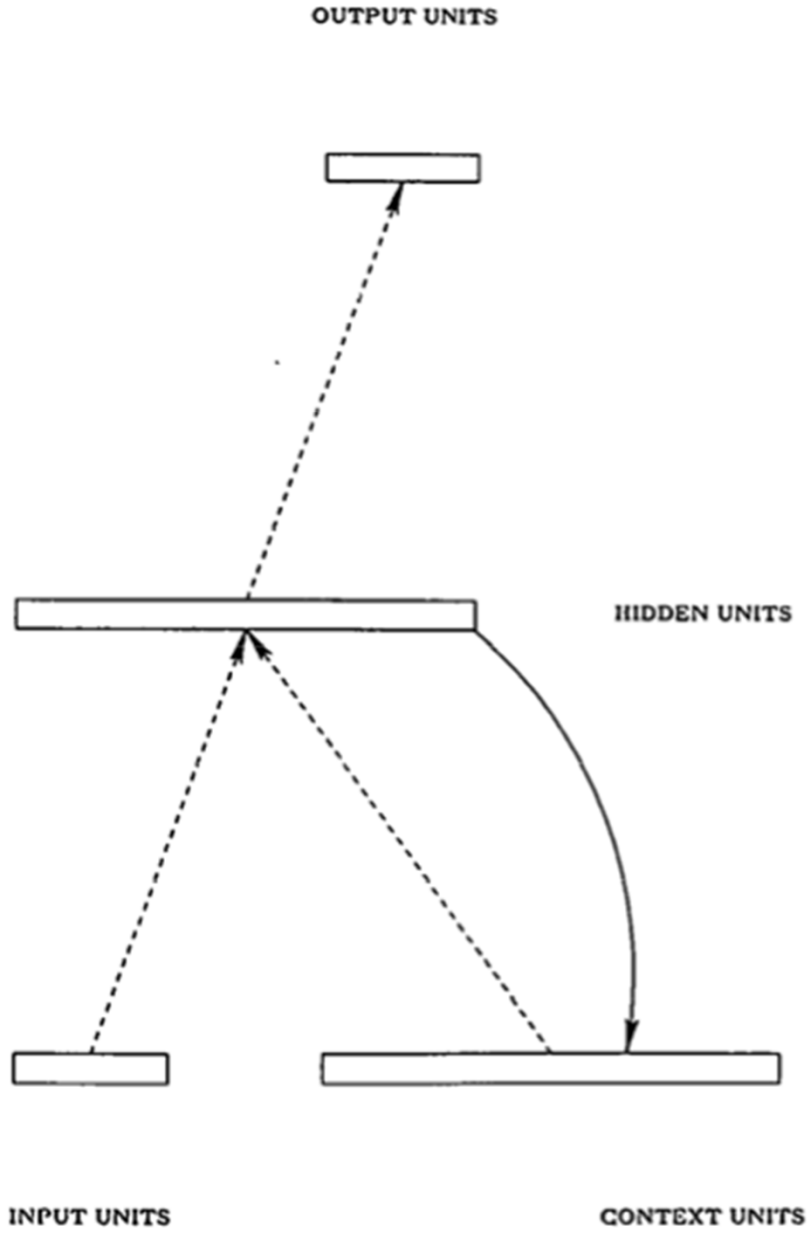


Figure 2. A simple recurrent network in which activations are copied from hidden layer to context layer on a one-for-one basis, with fixed weight of 1.0. Dotted lines represent trainable connections.

Elman (1990) built “simple recurrent networks” to capture different aspects of language performance in this framework.

All the model does is try to predict the next item it is going to see. It gradually learns to do this with experience.

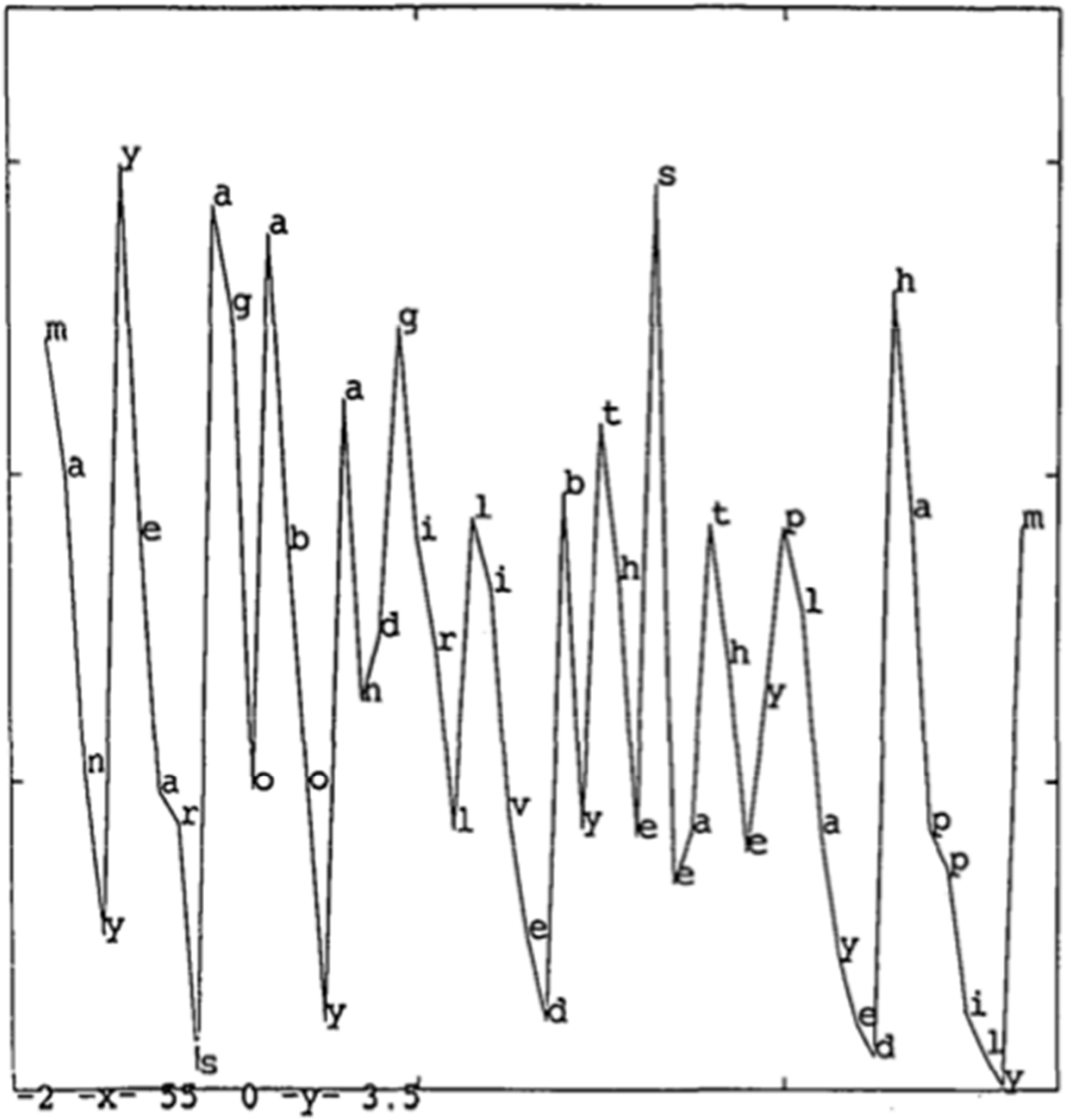


Figure 6. Graph of root mean squared error in letter-in-word precition task.

A letter-prediction model built on these principles can find words:

Manyyearsagoaboyandgirllived bytheseatheyplayedhappily…

becomes

“many years ago a boy and girl lived by the sea they played happily…”

Note prediction error (y-axis) goes down within words.

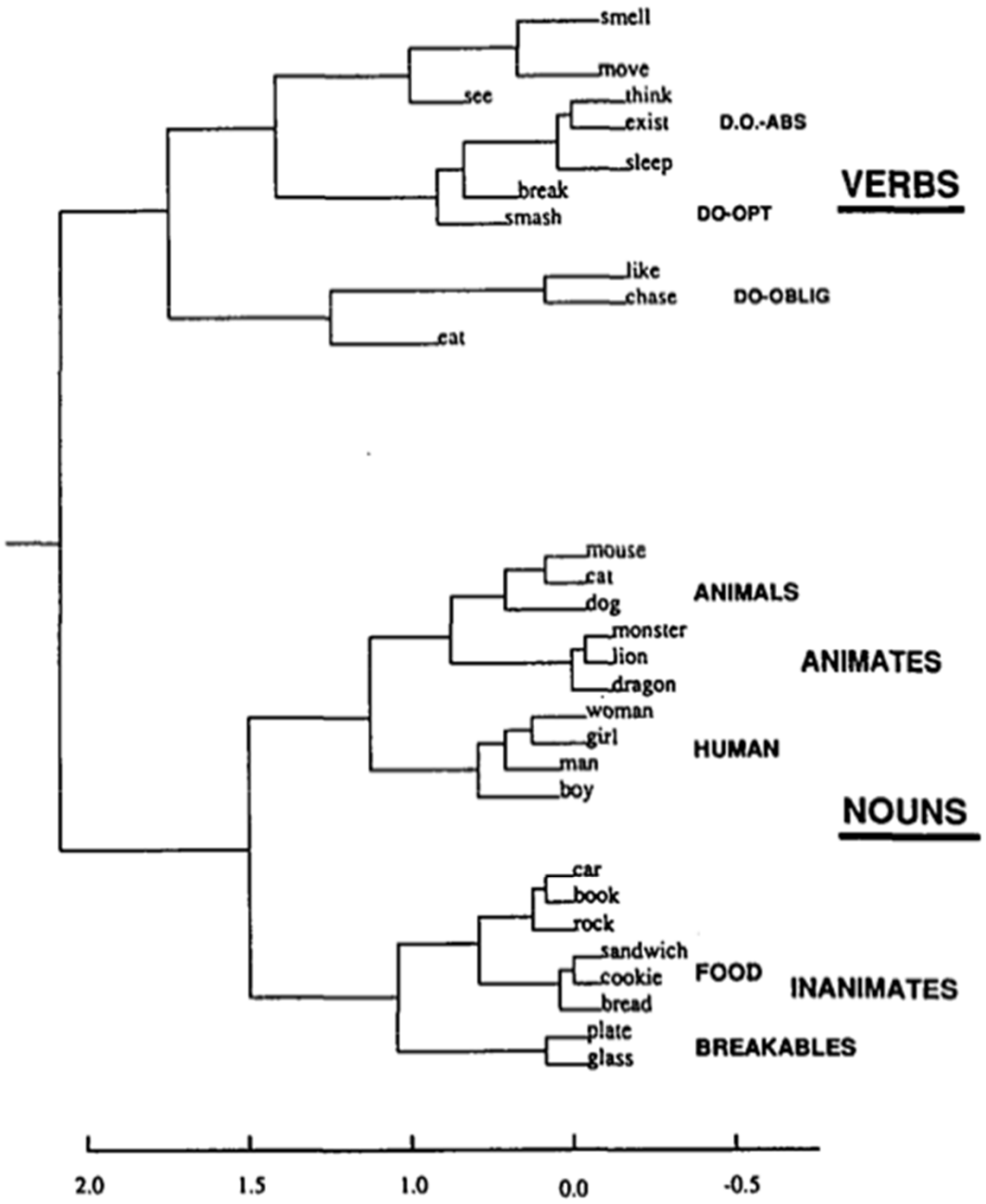


Figure 7. Hierarchical cluster diagram of hidden unit activation vectors in simple sentence prediction task. Labels indicate the inputs which produced the hidden unit vectors; inputs were presented in context, and the hidden unit vectors averaged across multiple contexts.

Train a similar model on simple sentences, and it starts to identify grammatical classes and even some semantic information, as a "side effect" of learning to predict the sequences.

...in fact this is how many engineering solutions (Natural Language Processing) work...

Massive corpora (collections of texts and transcriptions of spoken language) can be analyzed statistically to identify statistical patterns.

Chomsky appears to be correct that neither sentence appeared in the published literature before 1955. I'm not sure what he meant by "any of their parts," but certainly every two-word part had occurred, for example:

"It is neutral green, **colorless green,** like the glaucous water lying in a cellar." The Paris we remember, Elisabeth Finley Thomas (1942).

"To specify those **green ideas** is hardly necessary, but you may observe Mr. [D. H.] Lawrence in the role of the satiated aesthete." The New Republic: Volume 29 p. 184, William White (1922).

"**Ideas sleep** in books." Current Opinion: Volume 52, (1912).

But regardless of what is meant by "part," a statistically-trained finite-state model can in fact distinguish between these two sentences. Pereira (2001) showed that such a model, augmented with word categories and trained by expectation maximization on newspaper text, computes that **(a) is 200,000 times more probable than (b).**

From Peter Norvig's discussion of Chomsky's criti

USC Dornsife

Dana and David Dornsife
College of Letters, Arts and Sciences
Center for Computational Language Sciences

CCLS Speaker Series presents

LLM and human language: representations, judgments, and historical change



Adele E. Goldberg, Ph.D.

M. Taylor Pyne Professor of Psychology, Princeton

September 10th, 2pm

JOYCE J. CAMMILLERI HALL

In sum:
We've talked about two important traditions in psycholinguistics

1. GOFAI/Modular/Chomskyan:

Assume that linguistic knowledge is essentially *abstract* and *symbolic.*
Assume that linguistic representations are different in kind from other kinds of knowledge.
But all of this stuff has to interface with messy perceptual, motor, mnemonic, and other cognitive capa
The psycholinguist's job is to study this *interface*.

2.Emergentist/functionalist

Assume that symbol- and rule-like behavior are emergent properties of a statistical learning process.
Assume that perceptual, motor, and cognitive processes actually shape linguistic representations.
The psycholinguist's job is thus to study language use, and relate this to more general psychological ph

For Thursday

Read Menn, Chapter 3

Bring in SPEECH ERRORS to discuss next session! (Or email them to me so I can add them to slides.)

Consider as you read: Between the GOFAI and Emergentist traditions – where do Menn's sympathies lie?

Extra Stuff

Two movies:

Fast, Cheap, and Out of Control, by Errol Morris includes interviews with Rodney Brooks, who does emergentist robotics.

Is the Man who is Tall Happy? by Michel Gondry is essentially an animated interview with Chomsky about the concept of Universal Grammar.

An alternative approach to language: Emergentist/statistical learning approaches

The messiness demonstrated in the “i before e” rule is all over the place in every language:

phonetic categories are probabilistically defined over multiple dimensions

words are ambiguous in every way imaginable!

morphological paradigms like "add -ed" for the past tense in English have exceptions like RUN-RAN, SLEEP-SLEPT, etc.

syntactic rules are open to dialect/individual variation, innovation and change over time…

A problem with the GOFAI approach

Moravec's paradox : Proving theorems and solving geometry problems is comparatively easy for computers, but a supposedly simple task like recognizing a face or crossing a room without bumping into anything is extremely difficult.

The opposite is true, of course, for humans.

So maybe characterizing language in formal terms, as you would for a computer program, is a false trail?

(definition from Wikipedia)

Another problem: Fixed categories and rules are often “fuzzy” in practice.

This has, of course, been recognized for a long time:

66. Consider for example the proceedings that we call "games". I mean board-games, card-games, ball-games, Olympic games, and so on. What is common to them all?—Don't say: "There *must* be something common, or they would not be called 'games' "—but *look and see* whether there is anything common to all.—For if you look at them you will not see something that is common to *all*, but similarities, relationships, and a whole series of them at that. To repeat: don't think, but look!—Look for example at board-games, with their multifarious relationships. Now pass to card-games; here you find many correspondences with the first group, but many common

Wittgenstein (1953) *Philosophical Investigations.*

Many phenomena in language are probabilistic.

features drop out, and others appear. When we pass next to ball-games, much that is common is retained, but much is lost.—Are they all 'amusing'? Compare chess with noughts and crosses. Or is there always winning and losing, or competition between players? Think of patience. In ball games there is winning and losing; but when a child throws his ball at the wall and catches it again, this feature has disappeared. Look at the parts played by skill and luck; and at the difference between skill in chess and skill in tennis. Think now of games like ring-a-ring-a-roses; here is the element of amusement, but how many other characteristic features have disappeared! And we can go through the many, many other groups of games in the same way; can see how similarities crop up and disappear.

And the result of this examination is: we see a complicated network of similarities overlapping and criss-crossing: sometimes overall similarities, sometimes similarities of detail.

Wittgenstein (1953) *Philosophical Investigations.*

“Family resemblances,” similarity-based generalization, hunches

Psychologists like

Roger Shepard (perception)

Eleanor Rosch (categorization)

Daniel Kahneman and Amos Tversky (decision-making)

…and many others were producing results that made a lot of cognition seem less amenable to formal, rational interpretation.